

Kernel ERA Simulation — Results (working draft)

1 Results

We compared Kernel ERA with Linear ERA across a fully crossed factorial simulation. The predictive contrast is

$$\Delta R^2 = R_{\text{Kernel}}^2 - R_{\text{Linear}}^2, \quad (1)$$

and component recovery is measured by the difference in correlation between each estimated domain component and the component that generated the data,

$$\Delta \text{cor}_k = \text{cor}(\hat{f}_k^{\text{Kernel}}, f_k^{\text{true}}) - \text{cor}(\hat{f}_k^{\text{Linear}}, f_k^{\text{true}}). \quad (2)$$

Positive values of ΔR^2 and Δcor_k favor Kernel ERA.

Unless otherwise noted, all reported quantities are Monte Carlo means over 200 replications. Our primary analysis sample consists of *all operational outputs*, that is, every replication that completed successfully, whether or not the final Kernel ERA fit converged. To assess whether the conclusions depend on including these potentially unstable fits, Section 1.7 repeats the analyses on the subset of *kernel-stable replications*, defined as those in which the final fit converged and did not exceed the interpolation threshold.

1.1 Computational validity

The design comprised 24 conditions crossing the data-generating mechanism (linear, mixed, nonlinear), training size ($N \in \{150, 300\}$), per-set dimensionality ($P_k \in \{5, 20\}$), and target signal ($R^2 \in \{.30, .60\}$), with 200 Monte Carlo replications per condition, for a total of $24 \times 200 = 4,800$ condition–replication tasks.

All 4,800 tasks completed successfully at the computational level: there were no task-level errors, every expected selected-model summary was present, and every cross-validation (CV) candidate retained for selection satisfied the convergence-eligibility rule. Computational completeness is distinct from estimator convergence, however. Within these completed tasks, the final Kernel ERA fit failed to converge in a small number of replications, and it is this quantity—not the integrity of the computational pipeline—that motivates the two analysis samples defined above. We quantify it in Section 1.6, where it proves to be rare and confined to a small part of the design.

1.2 Predictive performance

The advantage of Kernel ERA over Linear ERA increased monotonically with the degree of nonlinearity in the data-generating mechanism. Averaged across conditions, ΔR^2 was +0.029 under linear truth, +0.151 under mixed truth, and +0.248 under nonlinear truth. The Kernel win rate—the proportion of replications in which Kernel ERA attained the higher out-of-sample R^2 —increased in the same order, from 0.56 to 0.94 to 0.99, as did the mean reduction in out-of-sample mean squared error (MSE) (Table 1). All three summaries follow the same ordering.

Figure 1 displays the same predictive contrast by training size, predictor-set dimensionality, target signal, and truth condition. The figure makes clear that the Kernel ERA advantage is small near the linear, low-dimensional settings, but increases under mixed and nonlinear truth. It also highlights the high-dimensional low-signal cases, where Kernel ERA improves generalization even when the data-generating mechanism is linear.

The scenario averages depend on P_k and R^2 in ways that are visible only at the condition level (Table 2; Figure 1). Two patterns are relevant.

First, under nonlinear truth the advantage is present in every condition. The eight nonlinear conditions yield ΔR^2 between +0.093 (condition 6) and +0.486 (condition 9), with Kernel win rates at or near 1.00 throughout.

Table 1: Predictive performance by data-generating mechanism (all operational outputs; means across the eight conditions within each scenario).

Truth	Mean ΔR^2	Kernel win rate	Mean MSE reduction
Linear	0.029	0.56	0.023
Mixed	0.151	0.94	0.159
Nonlinear	0.248	0.99	0.219

Note. $\Delta R^2 = R^2_{\text{Kernel}} - R^2_{\text{Linear}}$, evaluated on an independent test sample of 1,000 observations per replication. Positive values favor Kernel ERA.

Prediction advantage of Kernel ERA over Linear ERA

$$\Delta R^2 = R^2_{\text{Kernel}} - R^2_{\text{Linear}}$$

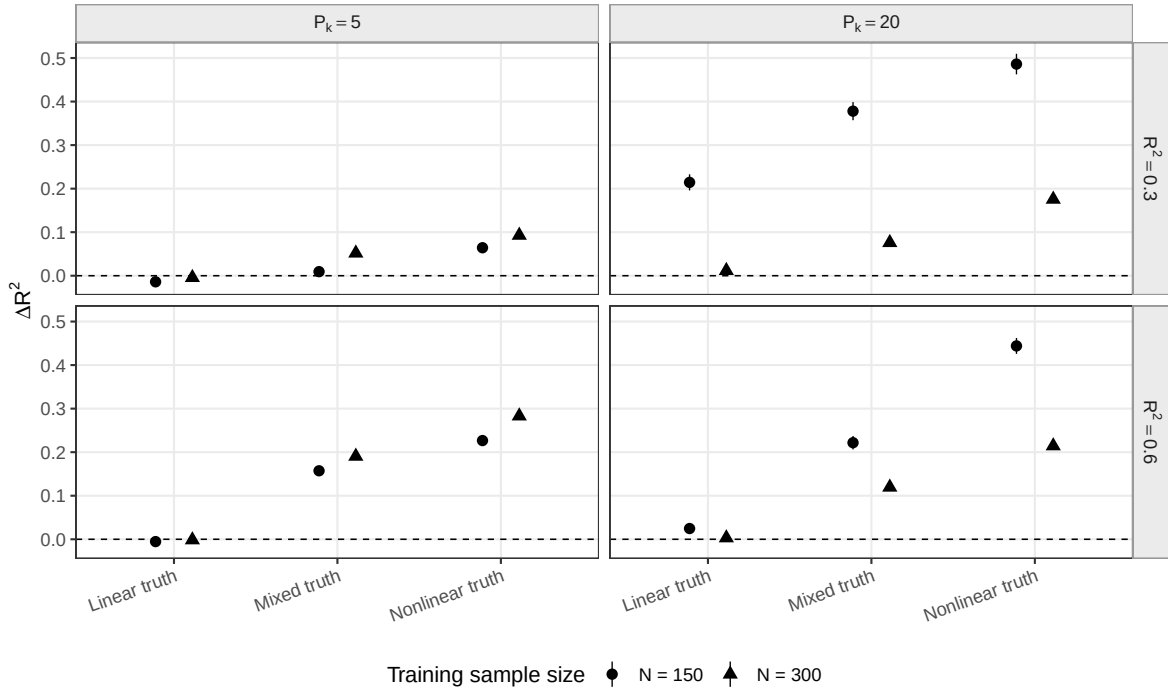


Figure 1: Prediction advantage of Kernel ERA over Linear ERA. Points show the condition-level Monte Carlo mean of $\Delta R^2 = R^2_{\text{Kernel}} - R^2_{\text{Linear}}$. Positive values favor Kernel ERA. Error bars show Monte Carlo uncertainty for the mean. The advantage increases from linear to mixed to nonlinear truth, with particularly large values in the high-dimensional, low-signal settings.

Second, the linear-truth conditions divide by dimensionality. In the low-dimensional linear conditions ($P_k = 5$; conditions 1, 4, 13, and 16), ΔR^2 is small and negative, ranging from -0.014 to -0.002 , with win rates below 0.50. This is the expected result: when the truth is linear and the problem is well conditioned, a nonlinear method has nothing to gain. In the high-dimensional linear conditions ($P_k = 20$), however, Kernel ERA can win by a wide margin. In condition 7 ($N = 150$, $P_k = 20$, $R^2 = .30$), $\Delta R^2 = +0.214$ and the win rate is 0.97, even though the generating mechanism is strictly linear. We return to this result in Section 1.5.

Table 2: Condition-level predictive performance (all operational outputs). Train-test gaps are the mean difference between in-sample and out-of-sample fit for each method.

Cond.	Truth	N	P_k	R^2	Mean ΔR^2	95% CI	Win	Train-test gap	
								Kernel	Linear
1	Linear	150	5	.30	-0.014	[-.018, -.010]	0.33	0.146	0.134
2	Mixed	150	5	.30	0.009	[.003, .015]	0.65	0.234	0.174
3	Nonlinear	150	5	.30	0.064	[.057, .071]	0.93	0.246	0.201
4	Linear	300	5	.30	-0.004	[-.006, -.003]	0.37	0.083	0.072
5	Mixed	300	5	.30	0.052	[.048, .056]	0.96	0.157	0.083
6	Nonlinear	300	5	.30	0.093	[.088, .098]	1.00	0.182	0.099
7	Linear	150	20	.30	0.214	[.196, .233]	0.97	0.227	0.744
8	Mixed	150	20	.30	0.378	[.357, .399]	1.00	0.242	0.893
9	Nonlinear	150	20	.30	0.486	[.462, .510]	1.00	0.267	0.971
10	Linear	300	20	.30	0.012	[.006, .017]	0.72	0.219	0.319
11	Mixed	300	20	.30	0.076	[.069, .083]	0.93	0.169	0.365
12	Nonlinear	300	20	.30	0.175	[.168, .183]	1.00	0.140	0.409
13	Linear	150	5	.60	-0.005	[-.008, -.003]	0.38	0.098	0.083
14	Mixed	150	5	.60	0.157	[.148, .166]	0.98	0.222	0.129
15	Nonlinear	150	5	.60	0.227	[.217, .236]	0.99	0.336	0.189
16	Linear	300	5	.60	-0.002	[-.002, -.001]	0.33	0.046	0.039
17	Mixed	300	5	.60	0.190	[.184, .197]	1.00	0.128	0.068
18	Nonlinear	300	5	.60	0.283	[.276, .290]	1.00	0.214	0.083
19	Linear	150	20	.60	0.025	[.015, .034]	0.74	0.333	0.430
20	Mixed	150	20	.60	0.222	[.206, .237]	0.97	0.335	0.673
21	Nonlinear	150	20	.60	0.444	[.426, .462]	1.00	0.305	0.859
22	Linear	300	20	.60	0.003	[.001, .005]	0.67	0.172	0.178
23	Mixed	300	20	.60	0.119	[.113, .126]	1.00	0.262	0.291
24	Nonlinear	300	20	.60	0.214	[.206, .223]	1.00	0.240	0.365

Note. “Win” is the Kernel R^2 win rate. Confidence intervals are Monte Carlo 95% intervals for the mean ΔR^2 . Condition 7 (shaded) is the linear-truth, high-dimensional, low-signal condition discussed in Section 1.5.

1.3 Two mechanisms underlying the Kernel ERA advantage

The predictive results show that Kernel ERA improves out-of-sample performance as the truth departs from linearity, **but they do not identify the reason**. Two mechanisms are possible, and they are not mutually exclusive.

The first is a **nonlinear representation advantage**. When the truth contains quadratic, interaction, or other nonlinear structure, the kernel feature map can recover domain components that a linear model cannot.

The second is a **regularized stabilization advantage**. In high-dimensional, low-signal settings, the RKHS ridge penalty and the convergence-eligible CV selection together yield a heavily regularized component estimator whose out-of-sample behavior is more stable than that of Linear ERA. **This mechanism can operate even when the truth is linear.**

The two mechanisms make different predictions about component recovery. Only the first should improve the correlation between estimated and true components; the second should improve prediction without

improving component recovery. We use this difference in Sections 1.4 and 1.5 to determine which mechanism is responsible for the gains observed in each part of the design.

1.4 Component recovery

Averaged over predictor sets, Δcor was -0.024 under linear truth, $+0.191$ under mixed truth, and $+0.285$ under nonlinear truth (Table 3). Under linear truth the value is negative, meaning that Linear ERA recovered the generating components slightly better than Kernel ERA did. This is the case even though Kernel ERA achieved a slightly higher mean ΔR^2 in the same scenario ($+0.029$; Table 1). Under mixed and nonlinear truth, component recovery favors Kernel ERA.

The difference in sign between prediction and component recovery under linear truth is informative. Because component recovery can improve only if the method captures the generating structure, and regularization alone cannot produce such an improvement, the predictive gains observed under linear truth cannot be attributed to the representation mechanism. They are consistent with the stabilization mechanism instead, which we examine directly in Section 1.5. Figure 2 summarizes the component-recovery pattern by functional form and predictor set. The figure shows that the Kernel advantage is not uniform across all components: it is absent under linear truth, negative for the linear set under mixed truth, and largest for the nonlinear sets, especially Set 2.

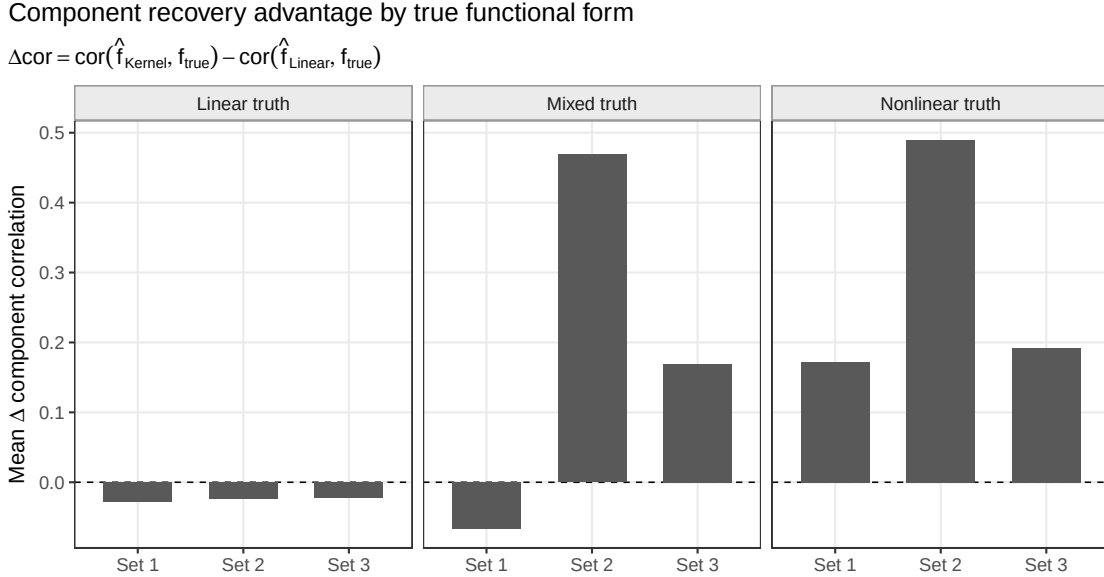


Figure 2: Component recovery advantage by true functional form. Bars show the mean difference in component correlation, $\Delta\text{cor} = \text{cor}(\hat{f}^{\text{Kernel}}, f^{\text{true}}) - \text{cor}(\hat{f}^{\text{Linear}}, f^{\text{true}})$, averaged across the design conditions within each truth condition. Positive values favor Kernel ERA. The largest gains occur for the nonlinear predictor sets, especially Set 2, which carries the strongest quadratic and interaction structure by construction.

Table 3: Component recovery by data-generating mechanism (all operational outputs; means across conditions).

Truth	Mean Δcor	Kernel component win rate
Linear	-0.024	0.43
Mixed	0.191	0.69
Nonlinear	0.285	0.95

Note. Δcor is defined in Eq. (2); positive values favor Kernel ERA. Win rates are averaged over predictor sets.

Breaking component recovery down by predictor set shows where the advantage is located (Table 4). Under mixed truth, in which some domains are linear and others are nonlinear, Kernel ERA does not improve recovery uniformly. It performs worse than Linear ERA on the linear predictor set (Set 1: $\Delta\text{cor} = -0.066$, win rate 0.21) and better on the two nonlinear sets, with the larger gain on the set carrying quadratic and interaction structure (Set 2: $\Delta\text{cor} = +0.469$, win rate 0.89) and a smaller gain on Set 3 (+0.169). Under nonlinear truth all three sets favor Kernel ERA, again with the largest value on Set 2 (+0.490).

Set 2 carries the strongest nonlinear structure by construction, and it is the set on which Kernel ERA improves recovery the most under both mixed and nonlinear truth. Recovery gains are therefore located in the parts of the model where nonlinear structure was introduced, and are absent or reversed where it was not.

Table 4: Component recovery by scenario and predictor set (all operational outputs).

Truth	Set	Mean Δcor	Win rate	Mean ΔRMSE
Linear	1	-0.027	0.39	0.033
	2	-0.023	0.45	0.025
	3	-0.022	0.46	0.021
Mixed	1	-0.066	0.21	0.099
	2	0.469	0.89	-0.236
	3	0.169	0.97	-0.135
Nonlinear	1	0.172	0.95	-0.137
	2	0.490	0.95	-0.222
	3	0.192	0.97	-0.142

Note. Set 2 carries the strongest quadratic and interaction structure by construction. ΔRMSE is the Kernel-minus-Linear difference in calibrated component RMSE, for which *negative* values favor Kernel ERA. Win rates refer to component correlation.

Component RMSE gives the same picture. Under mixed and nonlinear truth, ΔRMSE is negative on the nonlinear sets, reaching -0.236 on mixed Set 2, indicating lower error for Kernel ERA. Under linear truth, ΔRMSE is slightly positive on all three sets, indicating slightly lower error for Linear ERA. The detailed condition-level component pattern is shown in Supplementary Figure 5, where the same selective advantage for nonlinear components appears across sample size, dimensionality, and signal-strength conditions.

1.5 The linear-truth conditions and the stabilization mechanism

We now return to condition 7, in which Kernel ERA outperformed Linear ERA under a strictly linear truth ($\Delta R^2 = +0.214$; win rate 0.97). Section 1.4 showed that Kernel ERA has no component-recovery advantage under linear truth, so this predictive gain is not explained by the representation mechanism.

The train-test gaps in Table 2 indicate what is happening. In condition 7, the mean train-test gap for Linear ERA is 0.744, compared with 0.227 for Kernel ERA. Under the combination of small N , large P_k , and low R^2 , Linear ERA fits the training data closely but generalizes poorly, and its out-of-sample R^2 falls well below its in-sample fit. The RKHS penalty and the convergence-eligible CV selection keep Kernel ERA closer to its training performance. The same pattern appears, in weaker form, in the other high-dimensional linear conditions; in condition 10, for example, the gaps are 0.319 for Linear ERA and 0.219 for Kernel ERA, with $\Delta R^2 = +0.012$.

This is the stabilization mechanism described in Section 1.3, and it explains why the linear-truth mean ΔR^2 is slightly positive (+0.029) rather than zero or negative. The average is raised by the $P_k = 20$, $R^2 = .30$ conditions, while the low-dimensional linear conditions behave as expected, with small negative values of ΔR^2 .

Separating the two mechanisms accounts for the full set of predictive results. Gains under mixed and nonlinear truth are accompanied by improved recovery of the nonlinear components and are attributable to representation. Gains under high-dimensional linear truth are not accompanied by improved component recovery, are accompanied by a substantially smaller train-test gap, and are attributable to regularization.

1.6 Operating characteristics: stability and tuning

Final nonconvergence occurred in three of the 24 conditions: condition 7 (3.0% of replications; 6 of 200), condition 8 (1.5%; 3 of 200), and condition 11 (0.5%; 1 of 200). Across the full simulation, 10 of the 4,800 replications (0.2%) failed to converge. No condition showed any near-interpolation; the final interpolation rate was 0 in all 24 conditions, where interpolation was flagged at an in-sample fit above .98.

Aggregating over the design (Table 5), nonconvergence occurs in only one design cell, $N = 150$, $P_k = 20$, $R^2 = .30$, where the mean stable-fit rate is 0.985. At the same dimensionality but higher signal ($P_k = 20$, $R^2 = .60$), nonconvergence returns to zero. Dimensionality alone is therefore not the source of the instability; it arises from the combination of small N , large P_k , and low R^2 .

Table 5: Operational stability and tuning diagnostics by design cell (means over the three scenarios within each $N \times P_k \times R^2$ cell).

N	P_k	R^2	Stable rate	Nonconv. rate	Interp. rate	CV excl. rate
150	5	.30	1.00	0.00	0.00	0.08
300	5	.30	1.00	0.00	0.00	0.01
150	20	.30	0.99	0.02	0.00	0.64
300	20	.30	1.00	0.00	0.00	0.38
150	5	.60	1.00	0.00	0.00	0.01
300	5	.60	1.00	0.00	0.00	0.00
150	20	.60	1.00	0.00	0.00	0.27
300	20	.60	1.00	0.00	0.00	0.14

Note. “CV excl. rate” is the unrestricted-minimum exclusion rate: the proportion of replications in which the CV candidate with the lowest cross-validated error was excluded from selection because it failed the convergence-eligibility rule in at least one fold. The shaded row is the only cell with a nonzero nonconvergence rate.

Figure 3 presents the same operating characteristics as a compact design-cell summary. The estimator is stable across nearly all cells, whereas the CV-exclusion and tuning-boundary rates show where the tuning procedure was under the greatest pressure. In particular, the largest CV-exclusion rates occur in the $P_k = 20$, $R^2 = .30$ cells, especially when $N = 150$.

Two tuning diagnostics are worth reporting as operating characteristics of the method.

The first concerns the convergence-eligibility rule. The CV candidate with the lowest cross-validated error was excluded from selection in 64% of replications at $N = 150$, $P_k = 20$, $R^2 = .30$ and in 38% at $N = 300$, $P_k = 20$, $R^2 = .30$, compared with 8% or less in all other cells. In the most difficult conditions, then, many candidate solutions were unstable at the fold level, even though the eligible models that were ultimately selected produced stable final fits.

The second concerns the tuning grid. A boundary-concentration flag was raised in 17 of the 24 conditions, indicating that the selected (λ, σ) pair frequently fell on an edge of the grid. The direction of this behavior differs by condition. Under low-dimensional linear truth, the upper- σ boundary is selected most often, which corresponds to a smoother, closer-to-linear representation. Under the more difficult conditions, the upper- λ boundary is selected most often, which corresponds to stronger regularization. Both patterns are interpretable, but grid-boundary selection can still affect point estimates, and we therefore report a targeted sensitivity analysis on an expanded (λ, σ) grid for representative conditions in the Supplement.

Overall, the estimator was stable under the final tuning specification. Nonconvergence was limited to the most difficult low-signal, high-dimensional, small-sample conditions, and no condition showed near-interpolation.

1.7 Robustness to excluding nonstable fits

Because nonconvergent fits were rare and no replication interpolated, restricting the analyses to kernel-stable replications changed the results very little. The stable-only scenario means for ΔR^2 are effectively identical to the all-operational means: +0.0286 versus +0.0286 under linear truth, +0.1507 versus +0.1505 under mixed truth, and +0.2483 versus +0.2483 under nonlinear truth. The component-recovery summaries agree to the

Compact summary of stability and tuning pressure

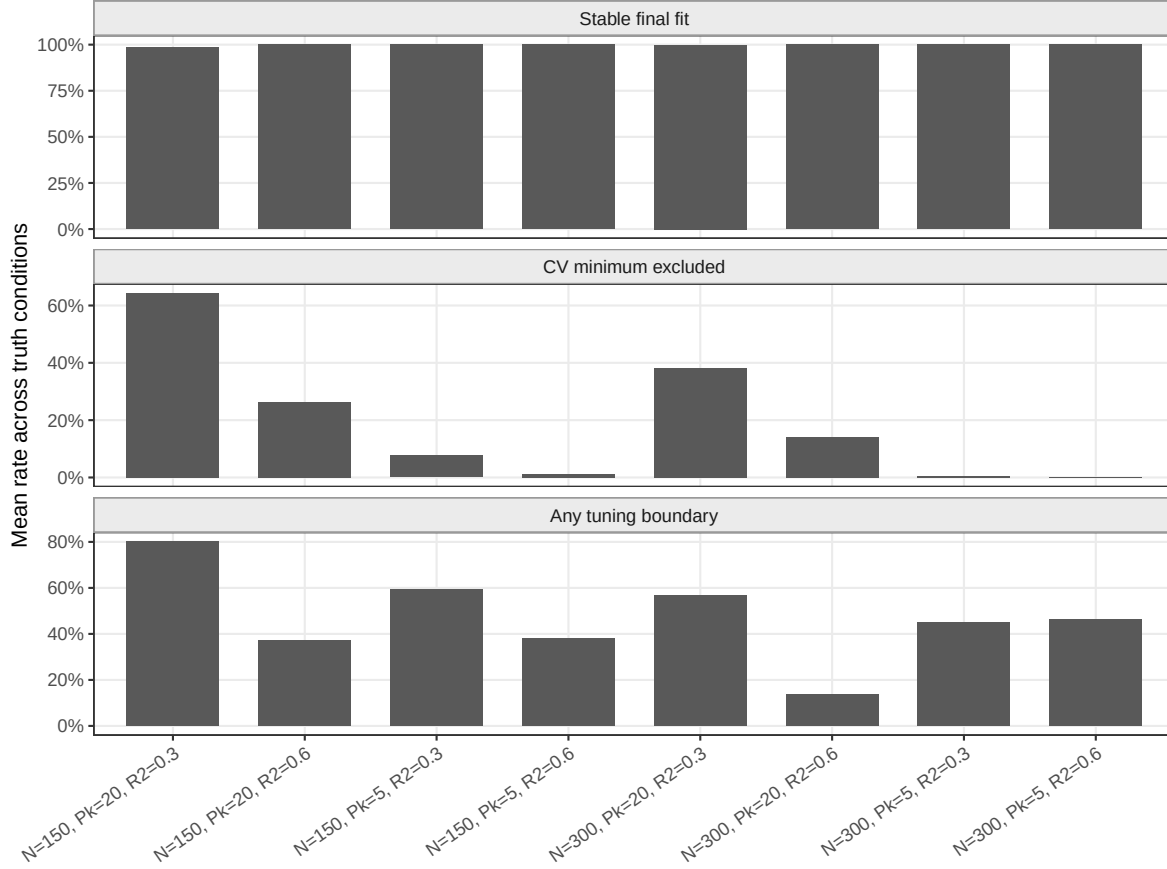


Figure 3: Operational stability and tuning diagnostics across design cells. Bars show mean rates across the three truth conditions within each $N \times P_k \times R^2$ design cell. The final fits are nearly always stable, whereas the unrestricted CV minimum is often excluded in the high-dimensional, low-signal cells. Boundary selection is common and is therefore treated as a tuning diagnostic rather than ignored.

fourth decimal place; for example, mixed Set 2 gives +0.4692 on all operational outputs and +0.4696 on kernel-stable replications. Only conditions 7, 8, and 11 lost any replications under the stable-only restriction, and each lost only a few. We therefore report the all-operational results as the primary summary of the method’s operating performance and treat the kernel-stable results as a sensitivity analysis that confirms these conclusions.

Figure 4 visualizes this robustness check. The points lie almost exactly on the diagonal, showing that the all-operational and kernel-stable summaries lead to the same condition-level conclusions.

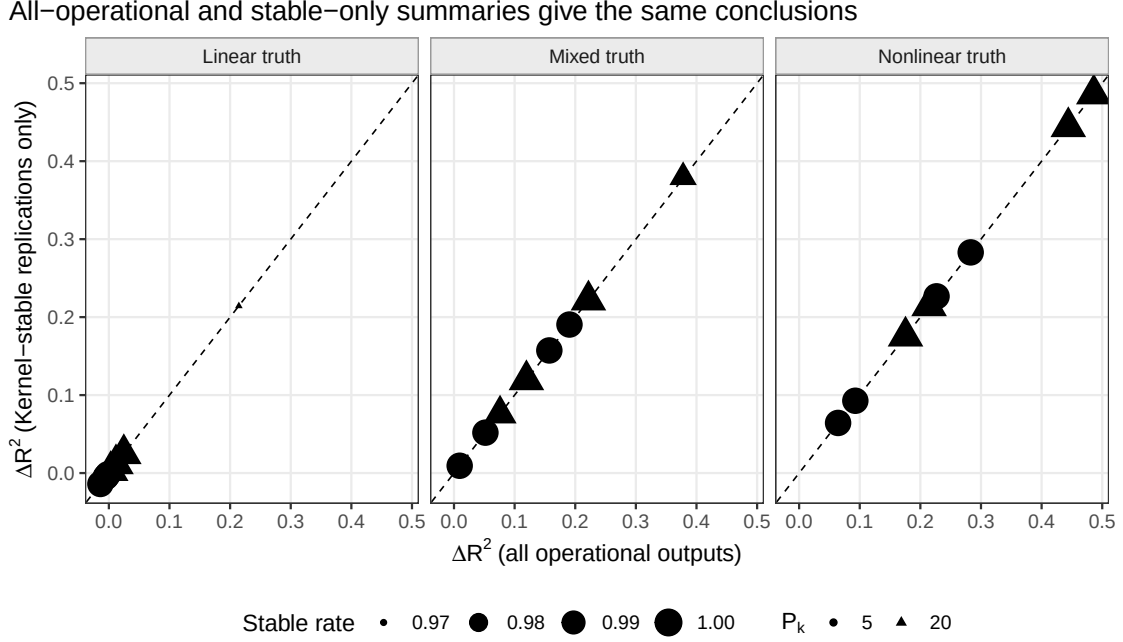


Figure 4: Comparison of all-operational and kernel-stable prediction summaries. Each point is one simulation condition. The horizontal axis gives ΔR^2 using all operational outputs, and the vertical axis gives ΔR^2 after restricting to replications in which the final Kernel ERA fit converged and did not exceed the interpolation threshold. The diagonal pattern shows that excluding the rare nonstable fits does not change the substantive conclusions.

1.8 Summary

First, the out-of-sample advantage of Kernel ERA over Linear ERA increases monotonically with the degree of nonlinearity in the data-generating mechanism, in ΔR^2 , in win rate, and in MSE reduction. **Second**, this advantage has two distinct sources. Under mixed and nonlinear truth, Kernel ERA recovers the true nonlinear domain components more accurately, with the largest gains on the predictor set carrying quadratic and interaction structure. Under high-dimensional, low-signal linear truth, Kernel ERA shows no component-recovery advantage but a substantially smaller train–test gap, indicating that its predictive gain there comes from regularization rather than from nonlinear representation. **Third**, estimation instability was limited. Nonconvergence occurred in 0.2% of replications overall and was confined to the most difficult design cell, and no replication interpolated. **Fourth**, the conclusions are unchanged when nonstable fits are excluded.

Supplementary figure candidates

Component recovery by design condition

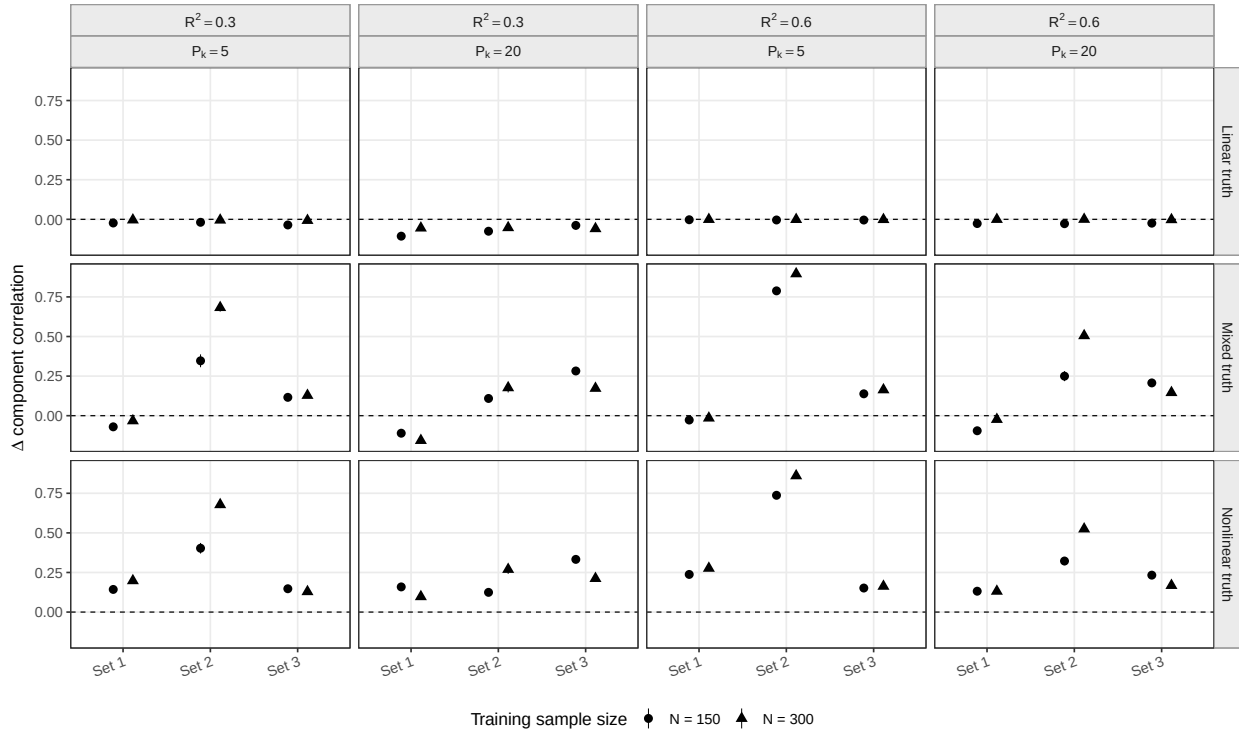


Figure 5: Supplementary Figure S1. Detailed component recovery by design condition. Points show condition-level mean differences in component correlation for each predictor set. This figure expands Figure 2 by showing how the component-recovery advantage varies by training size, predictor-set dimensionality, target signal, and truth condition.

Operating characteristics and tuning pressure

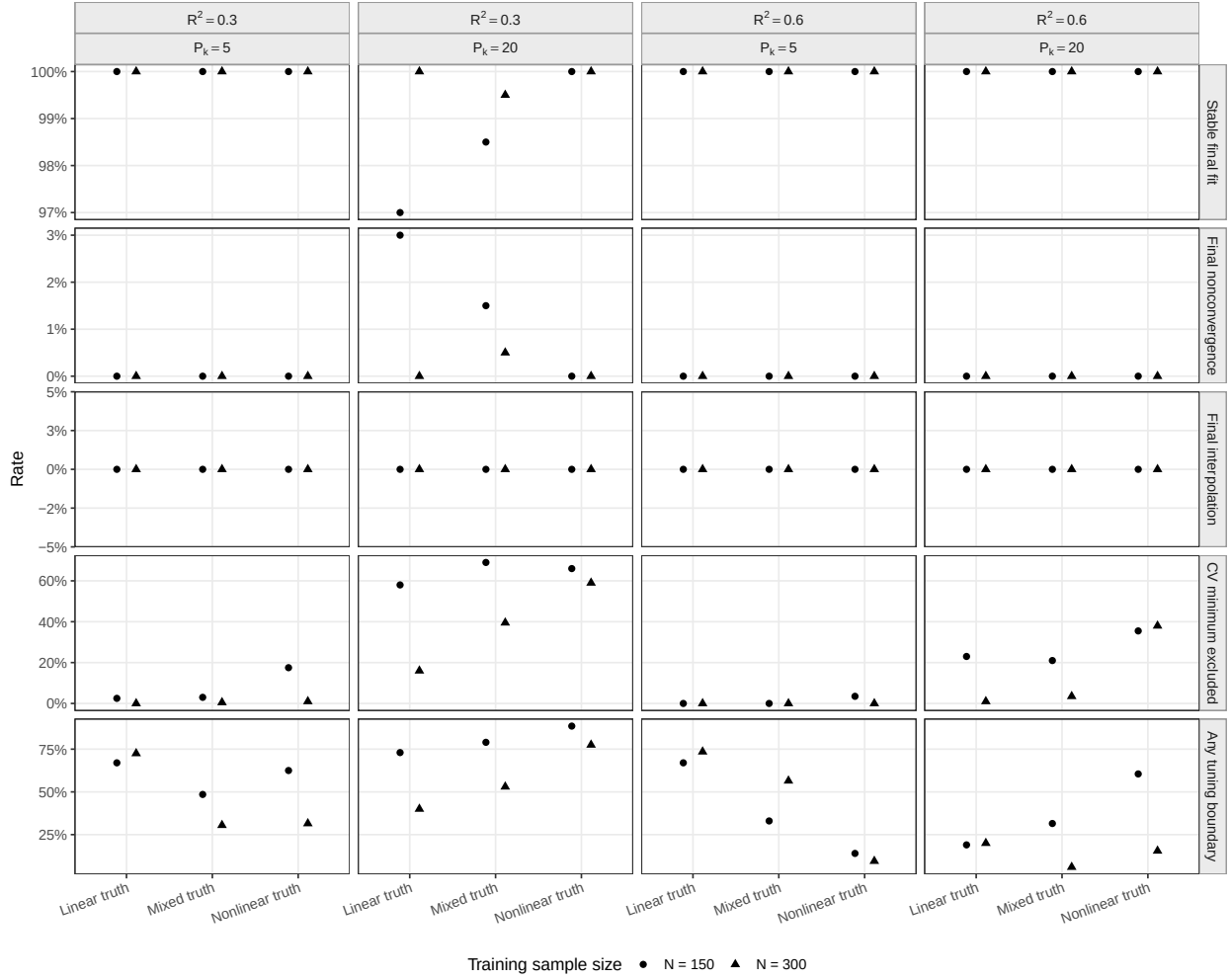


Figure 6: Supplementary Figure S2. Full operating characteristics and tuning pressure by design condition. The panels show stable-fit rates, final nonconvergence, final interpolation, exclusion of the unrestricted CV minimum, and boundary selection. Final interpolation is zero throughout, whereas CV-exclusion and boundary-selection rates identify the conditions in which tuning was most pressured.